EXPERIMENT NO. : 9 (MAIN)

DATE: 21/10/2026

1. AIM

To design and simulate a 6T SRAM cell using 180nm CMOS technology, determine the Cell Ratio (CR) and Pull-up Ratio (PR), and analyze the read and write operations through transient simulation.

2. TOOL

Cadence Virtuoso

3. THEORY

1. 6T SRAM Cell Architecture: A standard six-transistor static random access memory (6T SRAM) cell consists of two cross-coupled CMOS inverters forming a bistable latch that retains a single binary bit at complementary nodes (Q and $\overline{Q}$). Two NMOS access transistors connect the internal storage nodes to complementary bitlines (BL and $\overline{BL}$) under the control of a common Wordline (WL). The cross-coupled topology provides static data retention through positive regenerative feedback as long as the power supply is maintained.

2. Read Operation and Cell Ratio (CR): Prior to read access, complementary bitlines are precharged to V_{DD}. When the Wordline is asserted high, both access transistors turn on. For a cell storing logic '0' ($Q = 0\text{V}$), charge flows from the precharged bitline BL through the access transistor and the driver (pull-down) NMOS to ground, developing a sensing differential ΔV_{BL}. This creates a resistive voltage divider between the access transistor and the driver transistor. To guarantee read stability and avoid destructive read upsets, the voltage bump at node Q must remain strictly below the threshold voltage of the opposing driver NMOS ($V_{th,n}$). This requirement dictates that the driver transistor must be significantly stronger than the access transistor, defined by the Cell Ratio:
$CR = \frac{(W/L)_{\text{driver}}}{(W/L)_{\text{access}}} = \frac{W_n/L_n}{W_a/L_a} \geq 1.2.$

3. Write Operation and Pull-up Ratio (PR): During a write operation, the bitlines are driven by external write buffers to complementary logic states representing the input data, and WL is pulsed high. To successfully write a logic '0' into a node holding '1', the access transistor must overpower the pull-up PMOS transistor and pull the internal node voltage below the inverter switching threshold, triggering regenerative latch switching. To guarantee writeability, the pull-up PMOS must be weaker than the access NMOS, defined by the Pull-up Ratio: $PR = \frac{(W/L)_{\text{pullup}}}{(W/L)_{\text{access}}} = \frac{W_p/L_p}{W_a/L_a} \leq 0.6.$

4. Data Retention (Hold Mode): In standby mode, the Wordline is held at logic low ($WL = 0\text{V}$), turning off the access transistors and isolating the storage latch from the bitline capacitances. The positive feedback continuously maintains the latched complementary voltages with only minute static leakage currents.

4. TABLES

Table 4.4: Design & Simulation Parameters

Parameter	Value	Unit
Technology	180	nm
Supply Voltage (V_{DD})	1.8	V
Channel Length (L)	180	nm
Access Transistor Width (W_a)	800	nm
Driver NMOS Width (W_n)	1.6	μm
Pull-up PMOS Width (W_p)	400	nm
Bitline Capacitance (C_{BL})	50	fF
Simulation Stop Time	40	ns

Table 4.4: SRAM Sizing Ratios

Metric	Formula	Value	Status
Cell Ratio (CR)	W_n/W_a	2.00	Stable Read
Pull-up Ratio (PR)	W_p/W_a	0.50	Stable Write

Table 4.4: SRAM Operation & Transient Timing Analysis

Time Interval	Operation Mode	WL	BL	$\overline{BL}$	SWS	Node Q	Node $\overline{Q}$	Observation
0.0 to 2.7 ns	Initial Retention	1.8 V	0.0 V	1.8 V	1.8 V	0.0 V	1.8 V	Initial data '0' held stably
2.7 to 6.7 ns	Write '1' Operation	Pulsed High	1.8 V	0.0 V	1.8 V	0.0 V to 1.8 V	1.8 V to 0.0 V	Q flips to '1' upon WL pulse
6.7 to 10.7 ns	Write '0' Operation	Pulsed High	0.0 V	1.8 V	1.8 V	1.8 V to 0.0 V	0.0 V to 1.8 V	Q flips back to '0' upon WL pulse
10.7 to 12.0 ns	Precharge Phase	0.0 V	1.8 V	1.8 V	1.8 V to 0.0 V	0.0 V	1.8 V	Both bitlines precharged to 1.8 V
12.0 to 14.7 ns	Read '0' Operation	1.8 V	Discharges to 0 V	1.8 V	0.0 V	0.0 V	1.8 V	BL discharges to 0 V; stable non-destructive read
14.7 to 17.0 ns	Data Retention	0.0 V	0.0 V	1.8 V	0.0 V	0.0 V	1.8 V	Data '0' stably retained after read

5. CALCULATION

$$CR = W_n/W_a = 1.6/0.8 = 2$$

$$PR = W_p/W_a = 0.4/0.8 = 0.5$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

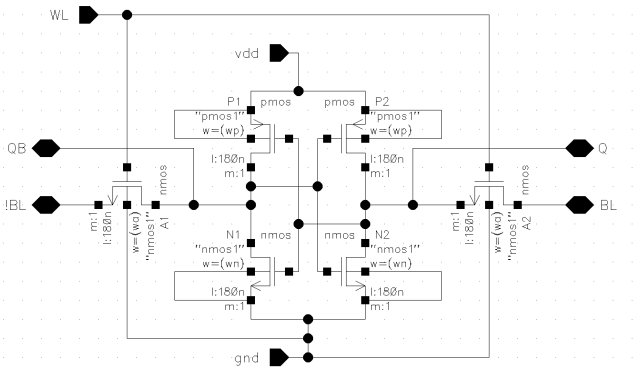


Figure 6.1: 6T SRAM cell schematic ($W_n = 1.6\,\mu\text{m}, W_p = 0.4\,\mu\text{m}, W_a = 0.8\,\mu\text{m}$)

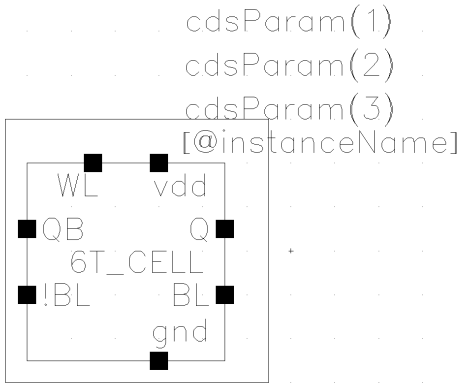


Figure 6.2: 6T SRAM cell symbol ($WL, BL, \overline{BL}, Q, \overline{Q}$)

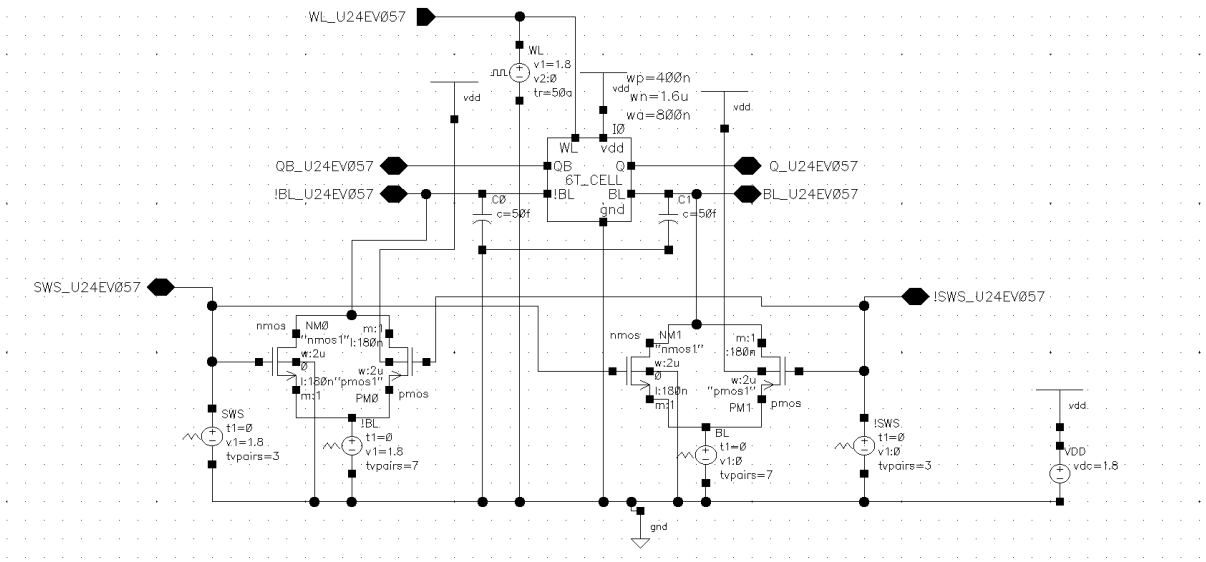


Figure 6.3: 6T SRAM cell testbench schematic ($C_{BL} = 50\text{fF}, V_{DD} = 1.8\text{V}$)

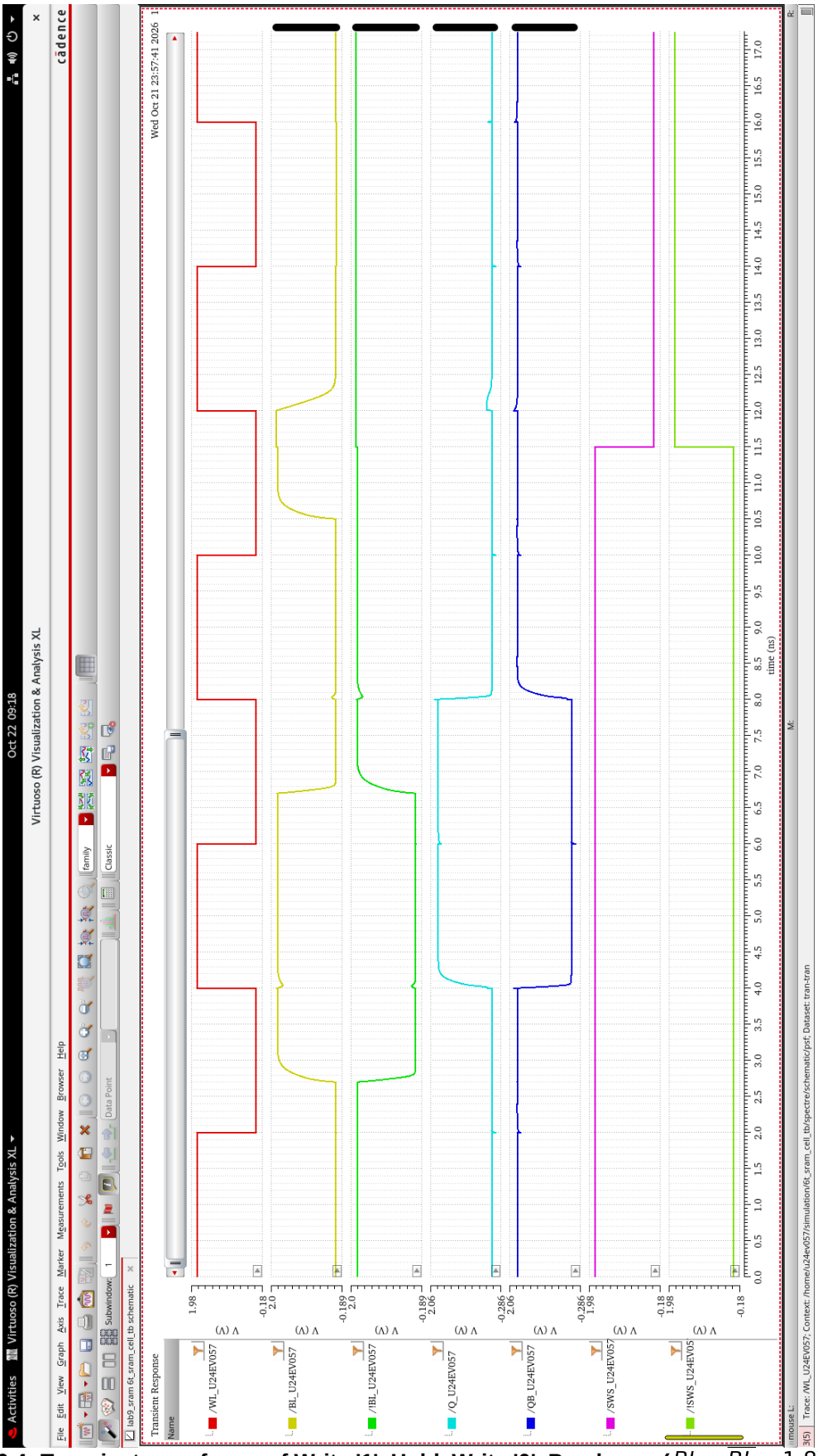


Figure 6.4: Transient waveforms of Write '1', Hold, Write '0', Precharge ($BL = \overline{BL} = 1.8\text{V}$), and Read '0'

Signals: WL (Wordline), $BL, \overline{BL}$ (Bitlines), $Q, \overline{Q}$ (Stored state), $SWS, \overline{SWS}$ (Write driver control)

7. CONCLUSION

1. The 6T SRAM cell was successfully designed and simulated in 180nm CMOS technology, achieving a Cell Ratio ($CR = 2.00$) for stable read ($CR \geq 1.2$) and a Pull-up Ratio ($PR = 0.50$) for stable write ($PR \leq 0.6$).

2. Transient simulations verified robust functionality across all states: successfully demonstrating Write '1', data retention, Write '0', precharge when both bitlines are $1.8V$, and non-destructive differential Read '0' where BL discharges to $0V$ without read disturbance.